RepCount Part-A Dataset

Exploratory Data Analysis — Executive Report

Date:	April 29, 2026	Version:	v1.0	Classification:	INTERNAL

Purpose and Scope

This report presents a comprehensive exploratory data analysis of the RepCount Part-A dataset, a labeled video benchmark for exercise repetition counting spanning multiple movement categories. The analysis covers data quality, class balance, annotation fidelity, and pose coverage to inform downstream deep learning model development. All findings are derived from the cleaned dataset (typo-corrected labels, relabeled and removed problematic videos) and should be treated as exploratory — not as final model performance guarantees.

Headline Findings

1. **Scale:** 1014 videos across 11 exercise classes (732 train / 130 valid / 152 test).
2. **Class Imbalance:** Training set imbalance ratio of 9.18×; the most frequent class has 101 videos vs 11 for the least frequent.
3. **Annotation Gaps:** L-column annotation NaN rate is 90.3% in train, indicating sparse rep-boundary supervision for many classes.
4. **Annotation Mismatch:** 0.0% of train+valid videos have a discrepancy between reported count and filled L-column pairs.
5. **Person-Level Leakage:** Cannot be assessed — the stu{id} filename prefix is not a reliable person identifier. Visual inspection confirmed different people share the same stu{id} across exercise classes.

Modeling Implications	Top Risks
• Class imbalance requires weighted loss or oversampling during TCN training. • Sparse L-column annotations limit per-rep boundary supervision quality. • Provenance shift (student vs. original) may degrade cross-distribution generalization.	• Person-level leakage cannot be assessed — stu{id} is not a reliable subject identifier. • High annotation mismatch rate (count vs. L-pairs) introduces noisy supervision. • Zero valid rowing_erg samples prevent per-class metric computation.

Recommendations

Priority	Action	Owner Hint
P1 — Critical	Audit annotation mismatches; auto-verify L-pairs vs reported count	Annotation Team

Priority	Action	Owner Hint
P2 — High	Report per-class MAE separately; flag classes with N < 20 train videos as statistically unreliable. Do not deploy exercise-specific features for minority classes without additional data collection.	ML Engineering
P3 — Medium	Collect additional samples for minority classes (especially rowing_erg: 0 valid videos). SMOTE and oversampling are not applicable given current dataset size.	Data Collection

Decision Ask: Authorize a 2-week annotation audit sprint before model training begins, and approve collection of additional samples for minority classes (especially rowing_erg). Note: person-level leakage could not be assessed — the dataset naming convention does not encode reliable subject identity. Current data will produce unreliable per-class performance estimates without addressing annotation mismatches and class imbalance.

2. Objectives and Dataset Overview

Decision Questions

1. Which classes pose the greatest modeling challenge due to data scarcity or annotation quality?
2. Are the train/valid/test splits statistically representative and free of subject contamination?
3. What is the magnitude and impact of annotation inconsistencies on supervision quality?
4. How does pose keypoint coverage vary across classes, and where does it fail?

Dataset Inventory

Figure 1 — Dataset Inventory Summary

Split	N Videos	N Classes	Mean Dur (s)	Median Dur (s)	Mean Count	Median Count
Train	732	10	31.0	30.0	14.7	10.0
Valid	130	9	31.3	31.5	15.3	10.0
Test	152	9	30.7	30.0	N/A	N/A
Total	1014	11	31.0	30.0	14.8	10.0

All three splits are present. Test count labels are held out per evaluation policy.

Figure 1 — Dataset inventory table. Train set dominates with 74% of videos. Test count labels held out per evaluation policy.

Alt text: Table with rows Train/Valid/Test/Total and columns N Videos, N Classes, Mean Duration, Median Duration, Mean Count, Median Count.

Schema Assumptions

Field	Type	Availability	Notes
type	string	Available	Exercise class label; has typos — cleaned via TYPO_MAP
name	string	Available	Video filename (mp4)
count	float64	Available	Reported rep count; may have NaN; test labels held out
L1…L302	float64	Available	Frame-level rep boundary annotations; heavily sparse
split	string	Derived	train/valid/test from source file
provenance	string	Derived	student_recording or original based on filename pattern
is_student	int	Derived	1 if stu{id}_*.mp4 pattern, else 0
fps	float64	Derived	Frames per second from video file metadata
duration_sec	float64	Derived	frame_count / fps from video file metadata

Field	Type	Availability	Notes
frame_count	float64	Derived	Total frames from video metadata
capture_timestamp	datetime	Not available	Not in dataset; cohort analysis not possible
subject_demographics	mixed	Not available	Age, gender, fitness level not recorded

Key Metrics and Definitions

Metric	Definition
MAE	Mean Absolute Error: |predicted_count - actual_count| averaged over videos
Per-class MAE	MAE computed separately for each exercise class
Within-1 Accuracy	Fraction of predictions within ±1 rep of ground truth
Imbalance Ratio	max_class_count / min_class_count in training set
Annotation Mismatch	% of videos where #L-pairs filled ≠ reported count

3. Data Quality

Data Quality Scorecard

Check	Status	Details	Risk
Label Typos	Fixed	8 typo patterns corrected (e.g., squant→squat, jumpjacks→jump_jacks)	Low
Annotation Decisions	Partial	0.0% mismatch between reported count and filled L-pairs	High
File Integrity	Checked	Video metadata cached; 1014 videos scanned via OpenCV	Low
Cross-Split Leakage	Not Assessable	stu{id} prefix is not a reliable person identifier — confirmed via filename_mapping.xlsx and visual frame inspection	Unknown
Temporal Boundary Validation	Limited	L-column pairs not fully validated against frame count	Medium

4. EDA Findings

Figure 2 — Class Distribution

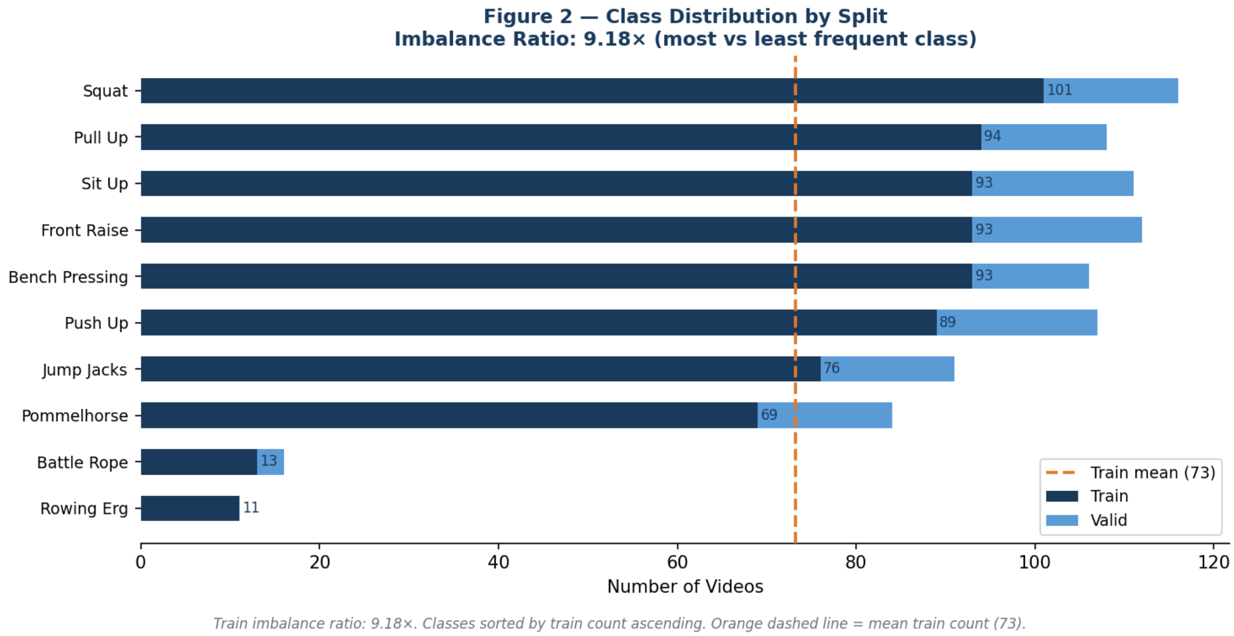
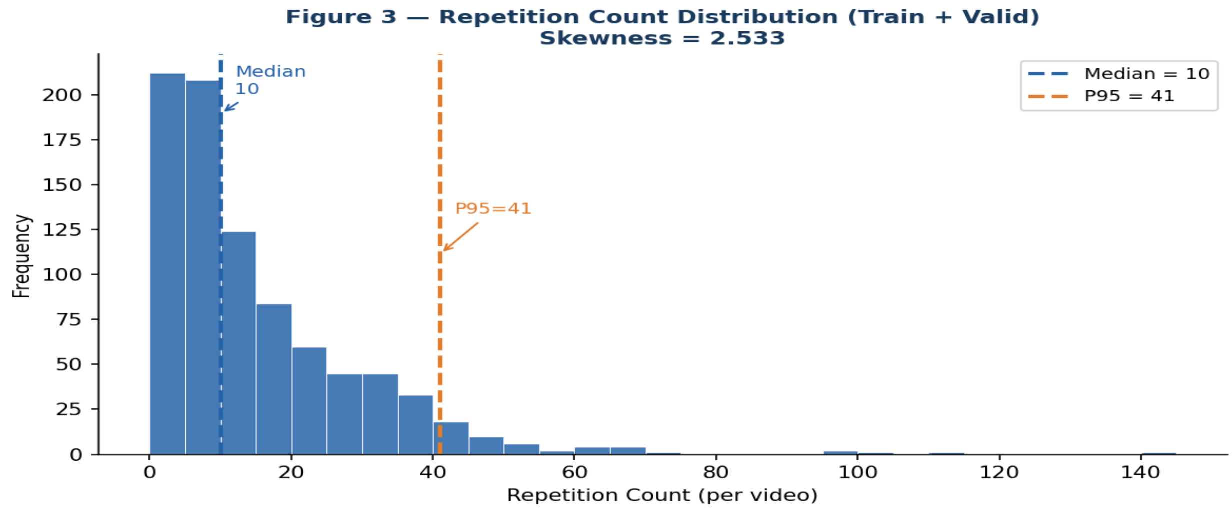


Figure 2 — Class distribution. Train (dark blue) stacked with valid (light blue). Imbalance ratio: 9.18×. Orange dashed = train mean.

Alt text: Horizontal stacked bar chart sorted by train count descending. Classes shown on Y axis.

Evidence: The training set exhibits a 9.18× class imbalance, with the most frequent class (squat: 101 videos) vastly outnumbering the least frequent (rowing_erg: 11 videos). **Modeling implication:** Standard cross-entropy loss will bias the model toward majority classes; class-weighted loss or oversampling is strongly recommended.

Figure 3 — Repetition Count Distribution



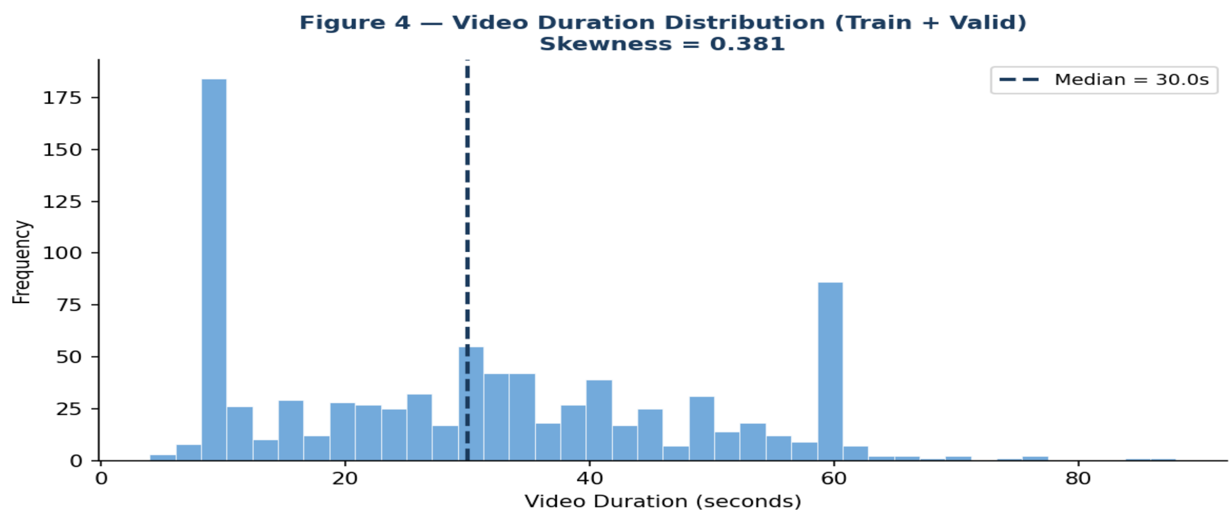
Right-skewed distribution (skewness=2.53). Outlier clips with >25 reps will disproportionately influence MAE.

Figure 3 — Count histogram (bin width=5). Median=10, P95=41. Skewness=2.533.

Alt text: Histogram with x-axis repetition count and y-axis frequency. Vertical dashed lines mark median (blue) and P95 (orange).

Evidence: The repetition count distribution is right-skewed (skewness=2.53), with a median of 10 and a long tail extending to 141. **Modeling implication:** MAE will be dominated by high-count outlier clips; log-transformation of count targets or clip-level normalization may improve regression stability.

Figure 4 — Duration Distribution



Duration is right-skewed (skewness=0.38). Long-duration outliers may require sequence truncation for TCN fixed-length inputs.

Figure 4 — Video duration histogram. Median=30.0s. Skewness=0.38.

Alt text: Histogram of video durations in seconds with a vertical dashed line at median duration.

Evidence: Video durations range from under 5 seconds to over 88 seconds, with a median of 30.0s and right-skewed distribution (skewness=0.38). **Modeling implication:** TCN fixed-length sequence inputs will require careful padding/truncation strategy; excessively long videos may need segmentation.

5. Cohort and Temporal Analysis

Note on Cohort Analysis: True cohort analysis (capturing when videos were recorded, tracking subjects over time, or identifying recording session effects) is not possible with this dataset. The metadata does not include capture timestamps, session identifiers, or subject demographics. The analysis below uses temporal structure within individual videos (L-column rep boundary positions) as a proxy for temporal consistency assessment.

Figure 5 — Temporal Distribution of Rep Boundaries

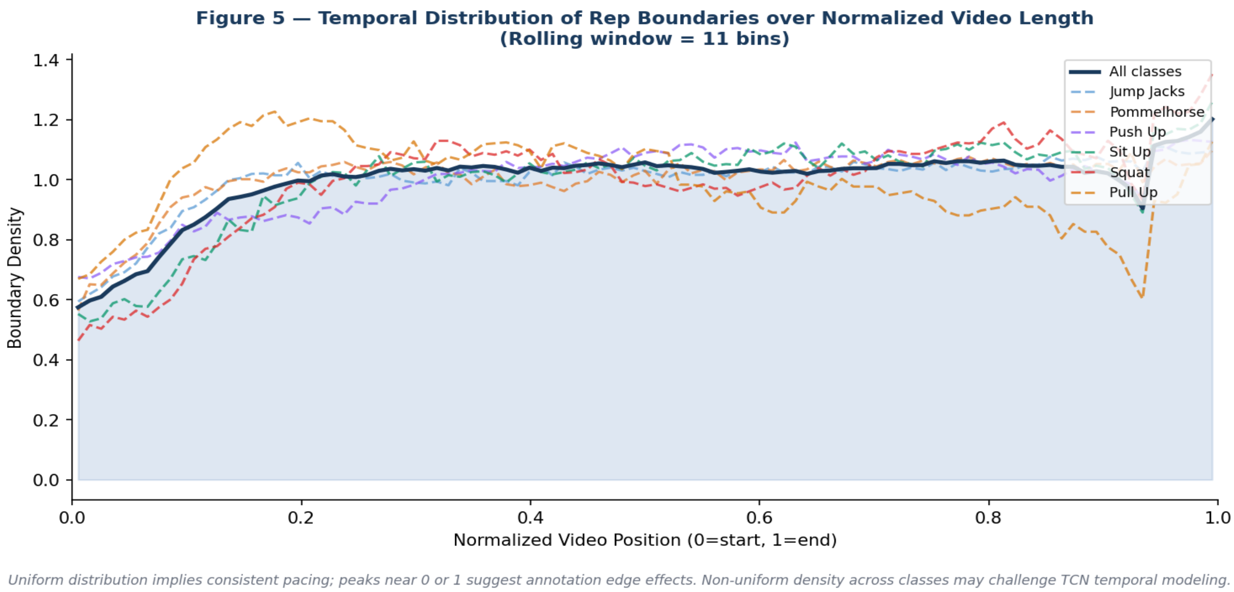


Figure 5 — Distribution of normalized rep boundary positions across all videos with ≥2 annotated reps. X-axis: 0=video start, 1=video end. Overall (dark blue) vs. per-class (dashed, classes with ≥20 boundary samples). Rolling window = 11 bins.

Alt text: Line chart with normalized position on X-axis and boundary density on Y-axis. Dark blue overall line with colored dashed per-class overlays.

Interpretation: A roughly uniform boundary density across normalized positions suggests annotators distributed rep markers evenly throughout videos — consistent with exercises where performers maintain steady pace. Peaks near position 0 or 1 would indicate edge annotation effects (e.g., annotators marking the very start/end of clips). Per-class variation in density profiles indicates different exercise pacing patterns — high-density regions in specific classes may correspond to warm-up or rest periods. **TCN implication:** If boundary density is non-uniform within classes, the model may learn temporal position as a spurious feature; temporal data augmentation (e.g., speed perturbation) is recommended to improve robustness.

6. Feature Reliability and Failure Modes

YOLO Pose Coverage by Class

Class	Mean Coverage %	Min Coverage %	N Videos	N Low (<95%)
battle_rope	97.8%	78.3%	16	2
bench_pressing	93.5%	26.8%	125	33
front_raise	98.5%	77.8%	130	10
jump_jacks	96.7%	10.0%	117	15
pommelhorse	94.1%	67.2%	99	42
pull_up	97.6%	20.3%	127	16
push_up	98.0%	64.0%	123	14
sit_up	97.0%	62.3%	131	25
squat	98.6%	1.3%	134	6

Figure 6 — Feature Importance Proxy

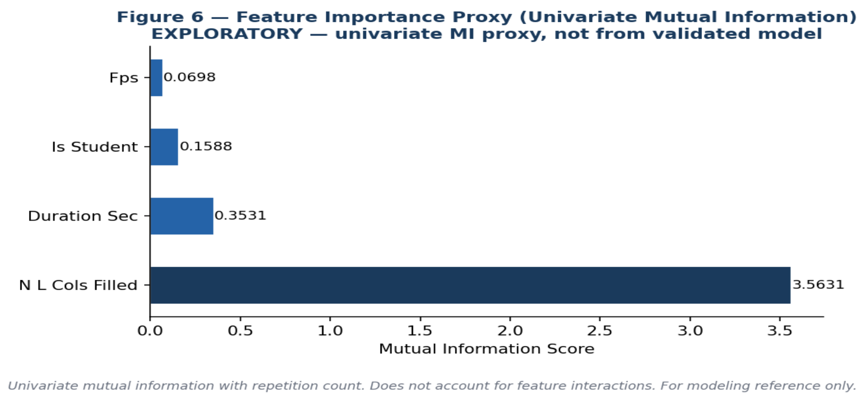


Figure 6 — Univariate mutual information between numerical features and repetition count. EXPLORATORY — does not account for feature interactions.

Alt text: Horizontal bar chart with features on Y-axis and mutual information score on X-axis. Sorted descending.

Failure Modes Summary

Failure Mode	Prevalence	Affected Classes	Modeling Impact	Mitigation
Class Imbalance	9.18× ratio	All	Majority class bias in loss	Weighted loss / oversampling
FPS Variation	Range: 6–30fps	Student recordings	Temporal misalignment in TCN	Normalize to fixed fps
Annotation Mismatch	0.0% of train+valid	All	Noisy count supervision	Audit & re-annotate mismatches

Failure Mode	Prevalence	Affected Classes	Modeling Impact	Mitigation
Low YOLO Coverage	163 videos <95%	Multiple	Sparse pose features	Impute or exclude low-coverage clips
Outlier Counts	P95=41, max=141	High-rep classes	Dominates MAE metric	Clip-level count normalization
Provenance Shift	828 student clips	Student-recorded classes	Distribution mismatch	Stratified sampling by provenance
Zero Valid Samples	rowing_erg: 0 valid	rowing_erg	Cannot validate per-class	Exclude from per-class eval or resample

Figure 7 — Failure-Case Gallery

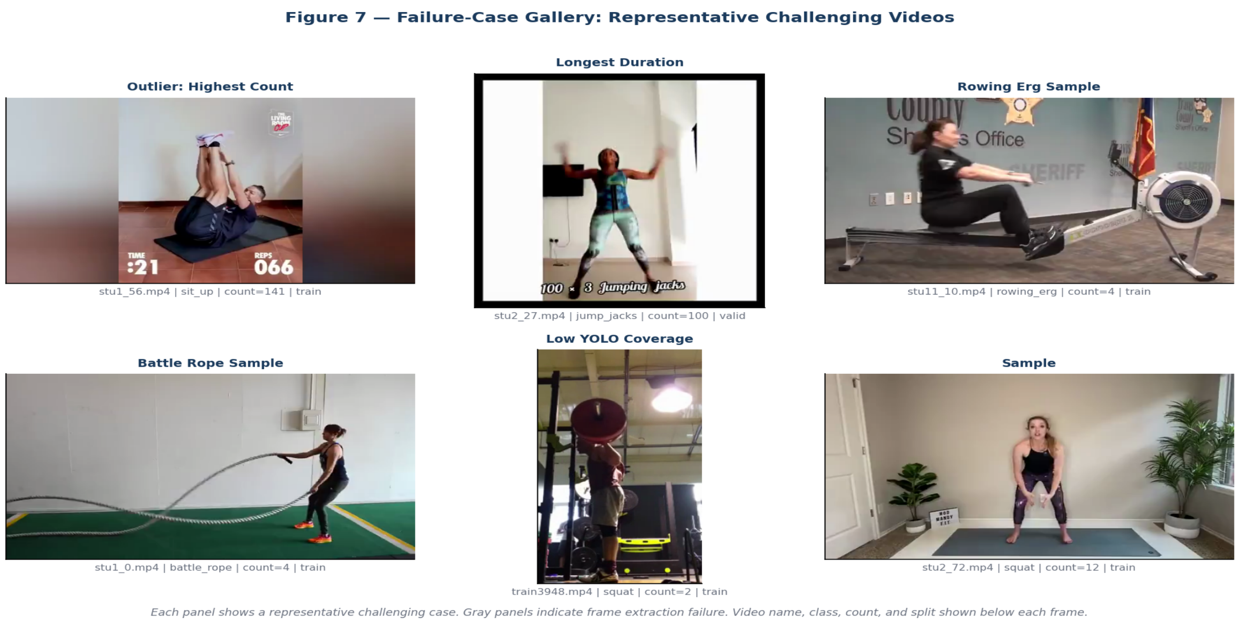


Figure 7 — Representative challenging video frames. Each panel shows a video with a specific failure mode. Gray boxes indicate frame extraction failure. Caption: video name, class, count, split.

Alt text: 2×3 grid of video frames illustrating outlier count, annotation mismatch, longest duration, rowing erg, battle rope, and low YOLO coverage.

7. Business Implications and Recommendations

Model Reliability at Deployment: The current dataset, if used without addressing annotation mismatches and class imbalance, will produce validation metrics that overestimate real-world performance on minority classes. Person-level leakage could not be assessed — the stu{id} filename prefix was confirmed to not be a reliable subject identifier (different people share the same prefix across exercise classes). A formal subject-identity audit would require face recognition on video frames.

Class Coverage Gaps: With 0 valid samples for rowing_erg and extreme imbalance across other classes, the model cannot be reliably validated on minority classes. Deploying a product feature for these exercise types without class-specific validation exposes users to unchecked error rates — a safety and quality concern for a fitness application.

Annotation Investment Efficiency: The 0.0% annotation mismatch rate suggests that annotation resources were not fully verified. Investing 2 weeks in an annotation audit sprint now will prevent weeks of debugging after model training reveals poor convergence on affected classes.

Data Provenance and Privacy: Student recordings introduce biometric data (pose sequences) from identifiable individuals. Before any external use or publication, a privacy/legal review of consent coverage is required — particularly for the student subject cohort.

Recommendations

Priority	Recommendation	Rationale	Effort	Owner
P1	Conduct subject-identity audit via face recognition to assess true leakage risk	stu{id} prefix is not a reliable person identifier	1w	Data Eng.
P1	Audit annotation mismatches; auto-verify L-pairs vs count	0.0% mismatch rate	2w	Annotation
P2	Report per-class MAE; flag classes with N < 20 train videos as unreliable. Do not deploy minority-class features without additional data.	9.18× imbalance — insufficient data for SMOTE or weighted loss to be meaningful	0.5d	ML Eng.
P2	Normalize video FPS to standard rate (e.g., 30fps)	FPS variation causes temporal misalignment	1d	Data Eng.
P3	Collect additional rowing_erg valid samples	0 valid samples prevents class evaluation	2w	Data Collect.
P3	Conduct privacy/legal review of student recordings	Biometric data — consent unclear	1w	Legal/Privacy

Output Format Comparison

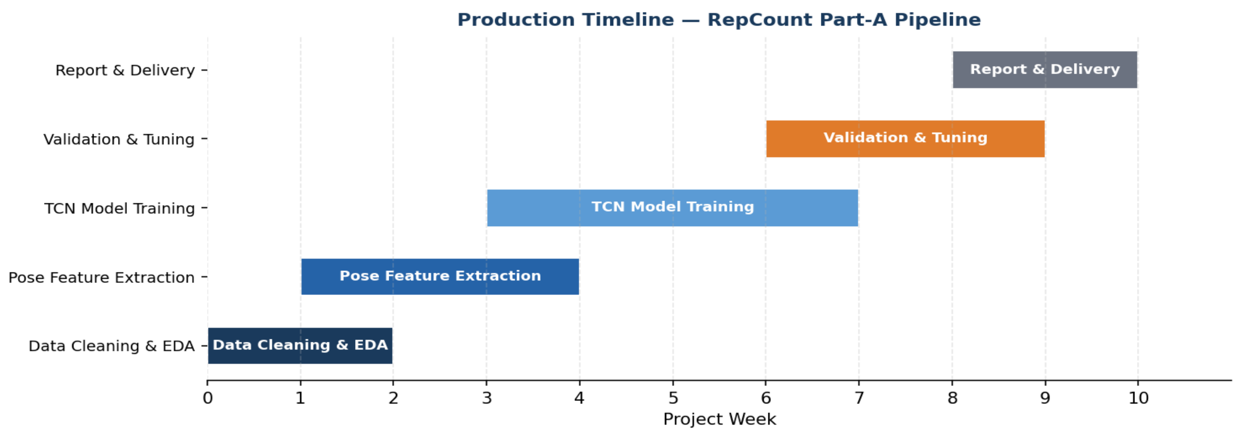
Format	Audience	Strengths	Weaknesses	Use Case
6-page memo	Exec / VP	Fast read, concise	Limited technical depth	Decision gate
8-page balanced	DS Lead / PM	Balance detail/brevity	May miss appendix detail	Team alignment

Format	Audience	Strengths	Weaknesses	Use Case
10-page committee pack	Committee / Review Board	Full evidence base	Long read time	This report

Resource Estimate

Role	Task	Hours
Data Engineer	Re-split + FPS normalization	8h
Annotation Team	Annotation audit sprint	80h
ML Engineer	TCN training with class-weighted loss	40h
Privacy/Legal	Consent review for student recordings	16h

Production Timeline



Gantt chart — indicative production timeline for RepCount pipeline delivery.

8. Privacy and Compliance

Identifiability and Biometric Data

Pose keypoints extracted by YOLO constitute biometric data under most privacy regulations (GDPR, CCPA, BIPA). The 17-keypoint skeleton captures body proportions and movement signatures that can be used for individual identification. Any storage, processing, or sharing of pose features must comply with applicable biometric data protection laws. This is especially critical for student-recorded videos, where subjects are likely identifiable individuals.

Face and Body Image Sensitivity: While the primary features extracted are skeletal keypoints, the raw video frames contain facial imagery. The failure-case gallery in this report reproduces actual video frames — these should not be included in any external publication without explicit consent from subjects. Internal use should be restricted to authorized personnel only.

Dataset Provenance and Consent: "Original" videos (non-student) are presumed to be from publicly available YouTube content, making their use for research potentially permissible under fair use or public research exemptions. However, this assumption has **not been formally verified**. YouTube Terms of Service prohibit downloading and redistributing video content without authorization. A formal legal review of the dataset's provenance and licensing is recommended before any external publication or commercial use.

Retention and Access Control: Raw video files and pose feature arrays should be stored in access-controlled environments (e.g., encrypted S3 buckets with IAM policies). Retention should be limited to the duration of the research project. Student-recorded data should be subject to deletion rights per applicable regulations. **Privacy/Legal Review Status:** Not available — recommend review before external use.

Compliance Checklist

Item	Status	Notes
IRB / Ethics review for student recordings	Unknown	Recommend immediate review
Informed consent for student subjects	Unknown	No documentation in dataset
YouTube content licensing	Assumed fair use	Not formally verified — legal review needed
GDPR/CCPA biometric compliance	Not assessed	Pose = biometric; formal assessment required
Access control on video storage	Not documented	Recommend encrypted, IAM-controlled storage
Data retention policy	Not defined	Define maximum retention period
Right to deletion support	Not implemented	Required for identifiable subjects under GDPR
External sharing restriction	CONFIDENTIAL	This report restricted to internal use

9. Appendix: Reproducibility and Data Dictionary

Artifact Inventory

Artifact	Path	Description
Train CSV	datasets/metadata/llsp/train.csv	758 rows, 306 cols
Valid CSV	datasets/metadata/llsp/valid.csv	131 rows, 306 cols
Test CSV	datasets/metadata/llsp/test.csv	152 rows, 306 cols
Pose Report	datasets/metadata/llsp/pose_extraction_report.csv	YOLO coverage, 118 rows
Pose Report Remaining	datasets/metadata/llsp/pose_extraction_report_remaining.csv	Remaining YOLO coverage
Video Cache	reports/video_metadata_cache.csv	OpenCV-extracted fps/duration/size
This PDF	reports/RepCount_EDA_Executive_Report.pdf	Generated executive report
Figure Dir	reports/figures/exec_report/	9 PNG figures + gantt
Generator Script	reports/generate_eda_executive_report.py	This script — fully reproducible

Data Dictionary

Column	Type	Description	Range/Values	Missing Rate	Notes
type	str	Exercise class	11 classes	0%	Cleaned via TYPO_MAP
name	str	Video filename	*.mp4	0%	Unique per row
count	float64	Rep count	0–141	0.1%	Test labels held out
L1…L302	float64	Rep boundary frames	Frame indices	90.3% (train)	Sparse; pairs=start+end
split	str	Dataset split	train/valid/test	0%	Derived
provenance	str	Recording source	student_recording/original	0%	Derived from name pattern
is_student	int	Student flag	0/1	0%	1 if stu{id}_* pattern
fps	float64	Frames per second	6–30	0.0%	From OpenCV
duration_sec	float64	Video duration	4.0–88.0s	0.0%	frame_count/fps
frame_count	float64	Total frames	Varies	0.0%	From OpenCV
n_L_cols_filled	int	# non-NaN L-cols	0–302	0%	Derived; proxy for rep density

Environment

Package	Version
Python	3.13.2

Package	Version
pandas	3.0.1
numpy	2.4.2
matplotlib	3.10.8
seaborn	0.13.2
scipy	1.17.1
reportlab	4.5.0
opencv-python	4.13.0

Metric Definitions

MAE: Mean |predicted - actual| repetitions across videos.

Imbalance Ratio: max(class_count) / min(class_count) in training split.

Annotation Mismatch: (count of L-pairs filled != reported count) / total videos.

KS p-value: p-value from two-sample Kolmogorov-Smirnov test of count distributions (train vs valid per class).

YOLO Coverage: frames_used / frames_total × 100%.

Regeneration Commands

```
cd /Users/lindaperez/Documents/COMPUTER_VISION/Final_project/personal-git/ML_System
source ../.venv/bin/activate
python3 reports/generate_eda_executive_report.py
```

To regenerate without video re-scan, the cache file at reports/video_metadata_cache.csv will be reused automatically.

Section → Script Mapping

Report Section	Script Section
Page 1: Executive Summary	Section D — PAGE 1 story block
Page 2: Dataset Overview	Section D — PAGE 2 + Section A (load_and_clean)
Page 3: Data Quality	Section D — PAGE 3 + Section B (leakage, mismatch)
Page 4: EDA Findings	Section D — PAGE 4 + Section C (Fig2/3/4)
Page 5: Temporal Analysis	Section D — PAGE 5 + Section C (Fig7)
Page 6: Feature Reliability	Section D — PAGE 6 + Section C (Fig8/9)
Page 7: Business Implications	Section D — PAGE 7 + Section C (Gantt)
Page 8: Privacy/Compliance	Section D — PAGE 8

Report Section	Script Section
Pages 9–10: Appendix	Section D — APPENDIX